

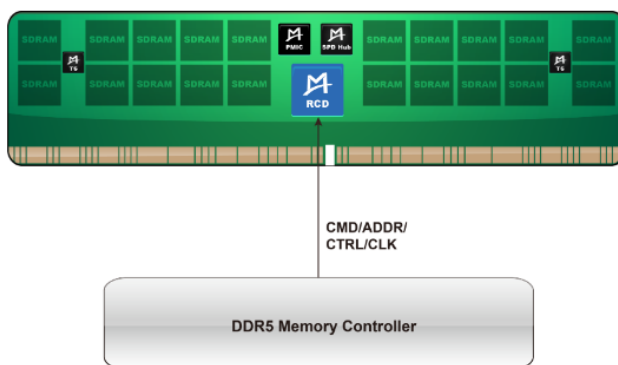
# **Exhibit CC-20**

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## M88DR5RCD01 (Gen1 DDR5 RCD)

### Product Description

M88DR5RCD01 is a JEDEC-compliant Gen1 DDR5 Registering Clock Driver (RCD), supporting data rates up to 4800MT/s and VDD supply of 1.1V. It is a two-channel RCD designed for driving addresses and controlling nets between the host controller and the DDR5 SDRAMs. The RCD can be used independently as a central buffer on an RDIMM or used in conjunction with Gen1 DDR5 data buffers (DBs) on an LRDIMM to enable higher capacity and lower power consumption for memory systems.



Notes: Here shows the application of RCD on the DDR5 RDIMM as an example.

### Feature

- Fully compliant with JEDEC DDR5RCD01 specification
- Speed up to 4800MT/s
- 1.1V VDD and 1.0V VDDIO voltages
- Two channel 1:2 registering buffer for address and control signals
- Integrated PLL clock driver distributing one differential clock pair to five differential pairs per channel
- Dual chip selects
- Parity checking across CA and DPAR separately on two sub-channels
- Output characteristics configurable through control registers
- CA, CS and DFE training modes and Dual Frequency support
- Sideband bus interface supporting I<sup>2</sup>C and I<sup>3</sup>C Basic modes
- Power saving modes: PDE power down, Self refresh w/wo Clock Stop modes
- Green package: 240-ball FCBGA

### Application

- DDR5 RDIMM
- DDR5 LRDIMM

### Document Download

 [M88DR5RCD01 Flyer](#)

DDR5 Memory Controller

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